

Software Requirements Specification (SRS)

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1. Introduction

Purpose:

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Artificial Medical Intelligence (A.M.I) system. It serves as a reference for developers, testers, and stakeholders to ensure alignment throughout the software development lifecycle.

Scope:

A.M.I is a web-based AI assistant designed to provide educational information related to personalized medicine while maintaining transparency through citations from peer-reviewed sources. The system will not diagnose illnesses or prescribe treatments.

Overview:

This document outlines system functionality, interfaces, constraints, performance expectations, and ethical considerations necessary for successful deployment.

2. General Description

Product Perspective:

A.M.I will operate as a standalone web application integrating a large language model, conversational interface, virtual avatar, and citation transparency features.

User Characteristics:

The system targets individuals seeking reliable health education. Users are expected to possess basic digital literacy.

Benefits:

- Increased trust through citation transparency
- Reduced misinformation
- Accessible health education
- Ethical AI interaction

3. Functional Requirements

- The system shall receive user prompts via a text-based interface.
- The system shall generate AI-driven responses using a trained large language model.
- The system shall prioritize personalized medicine and health-related topics.
- The system shall remind users that it is not a licensed medical professional.
- The system shall provide visible source citations.
- Users shall access the system through a web application.
- The system shall allow iterative dialogue through follow-up prompts.
- (Stretch) The system may support speech-to-text input.
- The system shall collect feedback after conversations.

4. Interface Requirements

- A virtual avatar will be displayed without obstructing the interface.
- Text-to-speech will generate synchronized speech and text responses.
- Audio controls (play/pause, volume, mute, replay) will be available.
- The avatar will communicate states such as loading, speaking, and idle.
- The avatar will maintain a friendly, professional medical assistant appearance.

5. Performance Requirements

- Responses shall be generated within 30 seconds under normal conditions.
- The system shall maintain at least 95% uptime.
- The system shall achieve a defined accuracy threshold during validation testing.

6. Design Constraints

- Backend development will primarily use Python with AI libraries such as TensorFlow, PyTorch, or Keras.
- Training may utilize NVIDIA GPU hardware or cloud services.
- The system must comply with privacy and data protection standards.

7. Non-Functional Attributes

- Responses must rely on validated, peer-reviewed medical sources.
- The system shall avoid storing PII/PHI unless encrypted.
- The application must be compatible with modern browsers.
- The system shall support iterative dialogue.

8. System Models – Use Case

Actors:

User – submits prompts, receives responses, listens to audio, submits follow-ups.

A.M.I – processes prompts, generates responses, provides citations, displays system states.

Primary Use Cases:

- Submit Prompt
- Process Prompt
- Generate Text Response
- Generate Source Citation
- Generate Audio Response
- Receive Text Response
- Submit Follow-Up Prompt
- Show System State (Loading, Speaking, Idle)

Exception Cases:

- Failed to Generate Response
- Failed to Receive Prompt

9. Preliminary Schedule

Requirements & Analysis: 3–4 weeks

System Design: 3 weeks

Model Development: 6–8 weeks

Testing: 3 weeks

Deployment: 2 weeks

Estimated Duration: 17–20 weeks

10. Appendices

Definitions:

LLM – Large Language Model

PII – Personally Identifiable Information

PHI – Protected Health Information

11. Conclusion

The development of the Artificial Medical Intelligence (A.M.I) system requires a comprehensive and well-structured Software Requirements Specification (SRS) to ensure clarity, alignment, and successful project execution. This document establishes a shared understanding among stakeholders, developers, testers, and project managers by clearly defining the system's functional and non-functional requirements. By outlining system behavior, constraints, performance expectations, and ethical considerations, the SRS serves as a foundational roadmap throughout the software development lifecycle. It supports informed decision-making, reduces ambiguity, and helps mitigate potential risks during design and implementation. Ultimately, this specification aims to guide the creation of a reliable, transparent, and user-centered AI medical assistant that prioritizes source credibility, privacy, and responsible AI practices. Adhering to the requirements detailed in this document will help ensure that A.M.I is delivered efficiently, meets stakeholder expectations, and achieves a high standard of software quality.